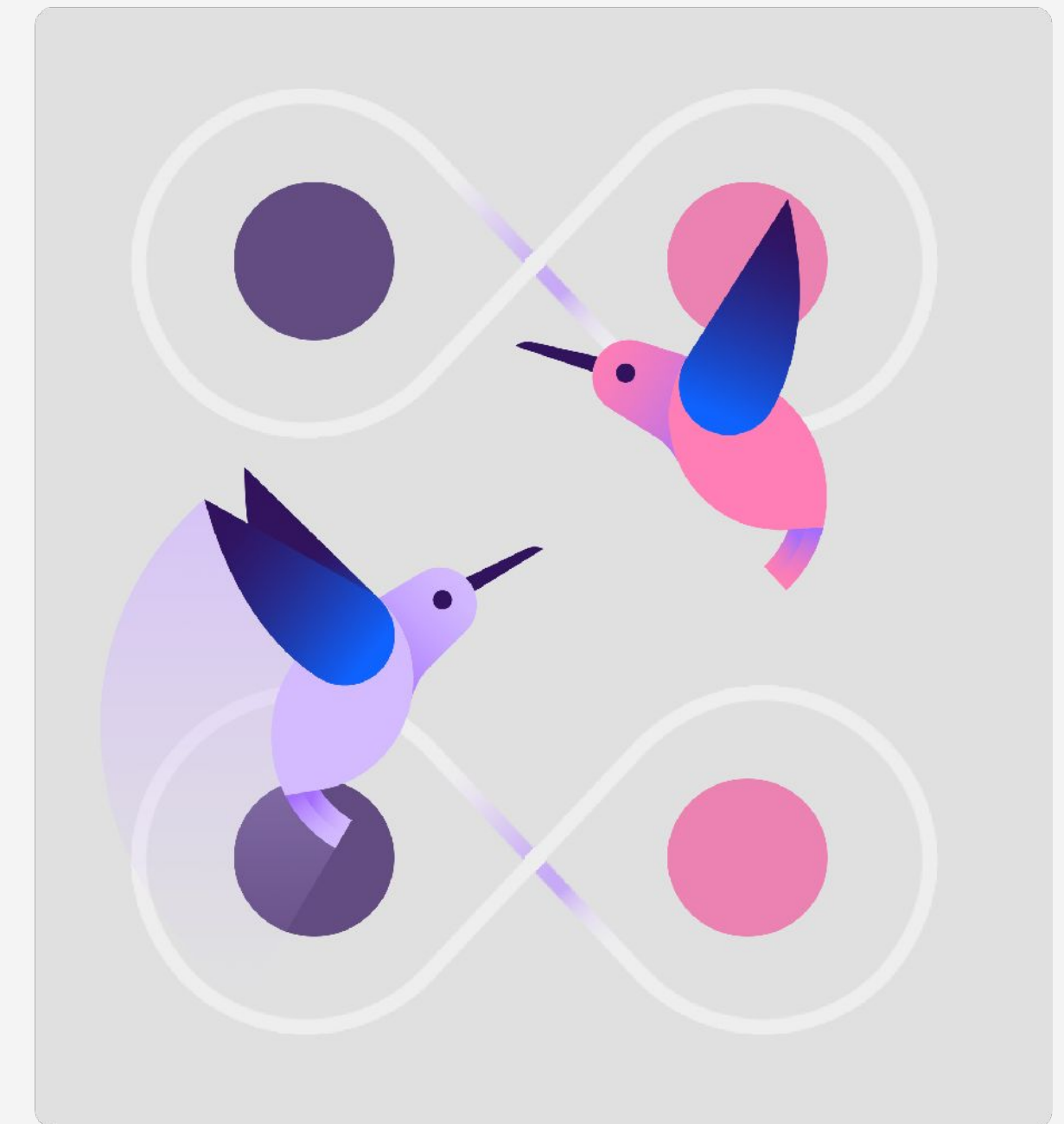
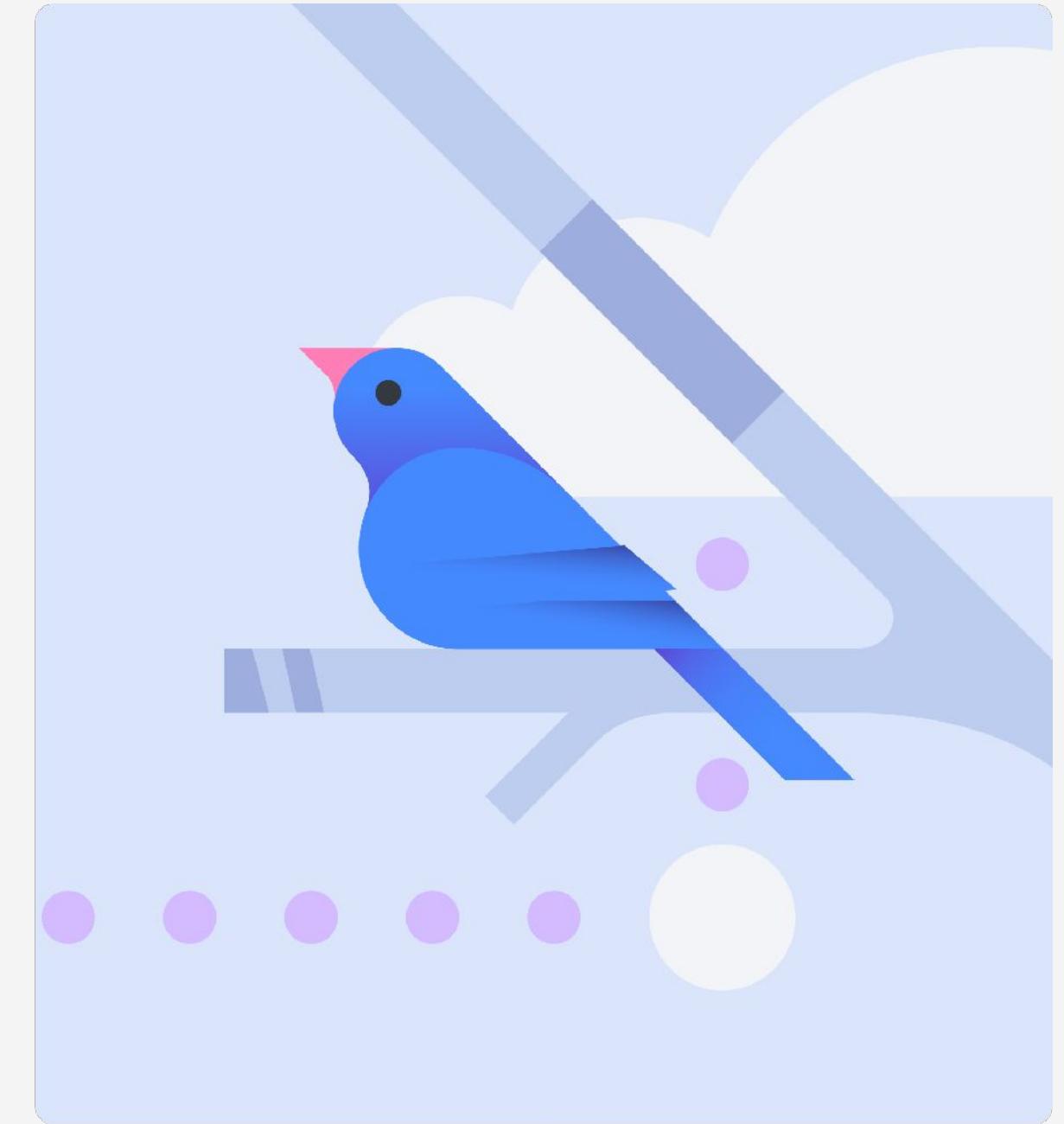
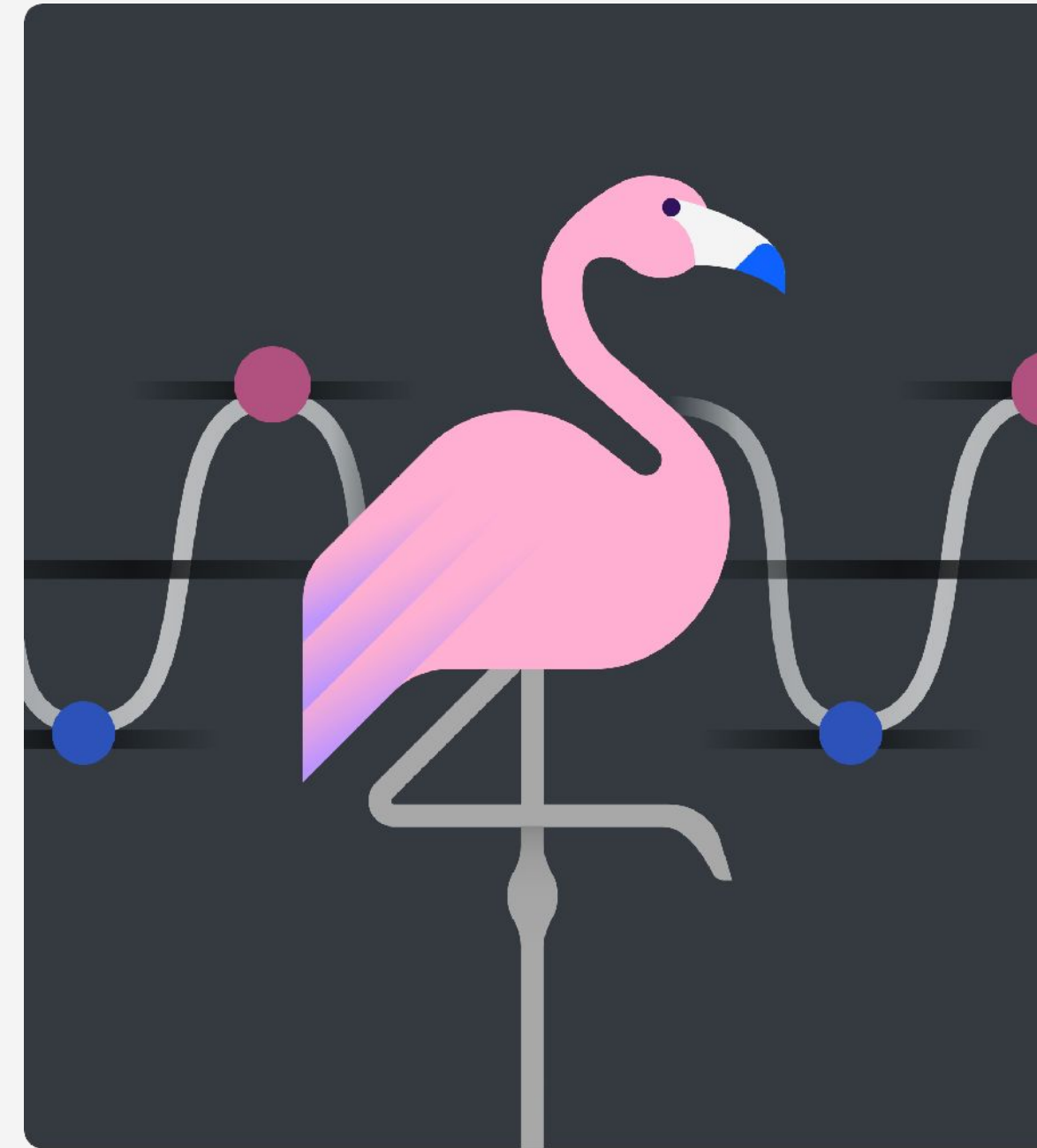


Single-qubit systems

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What you should be able to predict

- **Represent**

- $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
- Interpret α and β .
- Predict $P(0)$, $P(1)$.

- **Control**

- Reason about $R_x(\theta)$, $R_y(\theta)$, $R_z(\theta)$.
- Relate rotation angle to probability.

- **Transform**

- Explain X, Y, Z and H.
- Distinguish probability from phase.

- **Visualize**

- Interpret a statevector.
- Connect circuit $\rightarrow$ statevector $\rightarrow$ Bloch sphere.





A qubit is a state, not just a bit

The single-qubit state



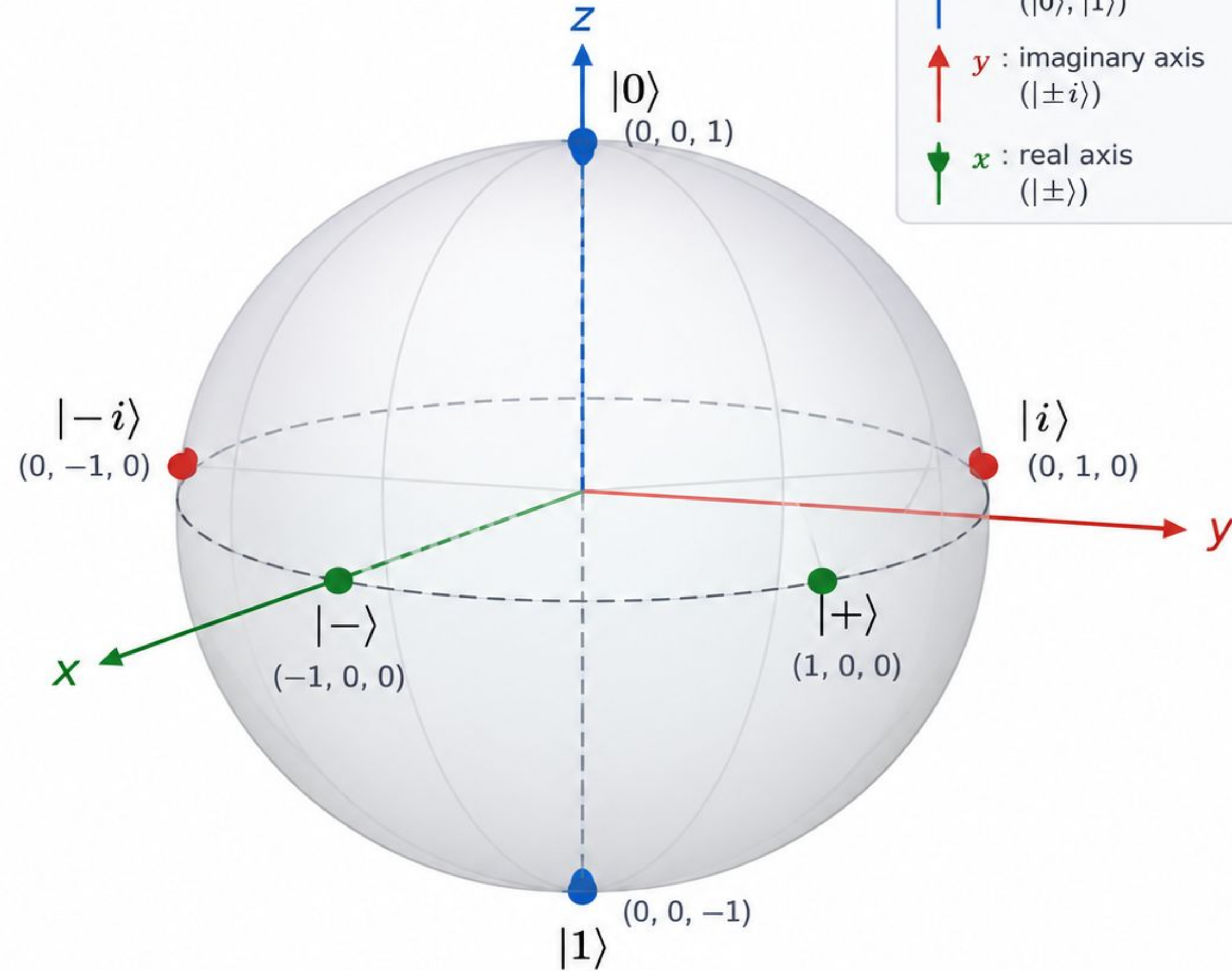
- **State**
 - $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
 - α and β are complex amplitudes.
 - Normalization
 - $|\alpha|^2 + |\beta|^2 = 1$.
- **Basis states**
 - $|0\rangle = [1, 0]^T$
 - $|1\rangle = [0, 1]^T$
- **Measurement**
 - $P(0) = |\alpha|^2$
 - $P(1) = |\beta|^2$
- **Amplitude $\neq$ probability.**

From notation to Qiskit

- Start in $|0\rangle$.
- Statevector $\rightarrow$ amplitudes.
- Measurement $\rightarrow$ classical outcomes.

Bloch Sphere

Single-qubit pure states on the surface of a sphere



$|+\rangle$ state

$$|+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

(1, 0, 0)

$|-\rangle$ state

$$|-\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

(-1, 0, 0)

$|i\rangle$ state

$$|i\rangle = \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle)$$

(0, 1, 0)

$|-i\rangle$ state

$$|-i\rangle = \frac{1}{\sqrt{2}} (|0\rangle - i|1\rangle)$$

(0, -1, 0)



Measurement:
from state to data

STATE

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Amplitudes + relative phase

MEASURE

$$P(0)=|\alpha|^2$$

$$P(1)=|\beta|^2$$

One shot $\rightarrow$ one outcome

REPEAT

Many shots $\rightarrow$ counts.

$$\text{frequency} = \text{count} / \text{shots}$$

Shot counts estimate probabilities; finite shots fluctuate around the ideal values.

Single-qubit gates

- **X**
- **Bit flip**

- $X|0\rangle = |1\rangle$
- $X|1\rangle = |0\rangle$

- **Y**
- **Bit + phase change**

- $Y|0\rangle = i|1\rangle$

Four gates you will use

- **Z**
- **Phase flip**

- $Z|0\rangle = |0\rangle$
- $Z|1\rangle = -|1\rangle$

- **H**
- **Superposition**

- $H|0\rangle = |+\rangle$
- $= (|0\rangle + |1\rangle)/\sqrt{2}$



Phase can hide in a histogram

- $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$
- $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$
- Both give $P(0)=P(1)=1/2$
- in the computational basis.

The relative sign is phase information.

Interference converts that phase difference into an observable population difference.

Rotations: continuous control



01 Rx(θ) rotation about X	02 Ry(θ) rotation about Y $Ry(θ) 0\rangle = \cos(θ/2) 0\rangle + \sin(θ/2) 1\rangle$	03 Rz(θ) rotation about Z changes relative phase	04 Predict θ changes amplitudes → probabilities $θ=π/3 \Rightarrow P(0)=0.75, P(1)=0.25$
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Reading the statevector in Qiskit

```
from qiskit.quantum_info import  
Statevector
```

```
state =  
Statevector.from_instruction(qc)  
state.data →  $[\alpha, \beta]$ 
```

```
state.probabilities_dict()  
→ {"0":  $|\alpha|^2$ , "1":  $|\beta|^2$ }
```

Predict first. Then verify.

$|+\rangle \rightarrow [1/\sqrt{2}, 1/\sqrt{2}]$

$|-\rangle \rightarrow [1/\sqrt{2}, -1/\sqrt{2}]$

The sign is phase information.

Statevector

$[\alpha]$

$[\beta]$

probabilities_dict()

exact ideal probabilities
(no shots)





Predict → build → visualize → explain

Quantum circuits manipulate amplitudes and phase.

Measurement reveals probabilities.



Ready for the notebook?

Predict X , H , $H \rightarrow Z$, and $R_y(\pi/3)$
before executing the code.